Promo Finder Finding ad-ready promos in microdrama screenplays

Kuku AI Generalist Assignment · Problem 1: Find the Promos (Ads)

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PROTOTYPE GitHub RAW OUTPUTS Google drive (both runs)

50	14,291	25	~12 s · $0
episodes read	candidate clips scored	ranked promos out (21 episodes)	per show, rules only

1 Why Problem 1, and how I scoped it

Promos are how a microdrama gets its viewers. Kuku ships a new show every week, and today an editor has to watch every episode to find them, so this one step caps how many shows can be promoted. It's also the only problem with real inputs I could check my output against. I scoped it to the brief's literal job: **find the moments and their start/end points**, then rank them. No editing. The output is an editor's shortlist, not the final ad.

Assumptions I made where the brief was silent

- Screenplay timestamps match the final cut. Clips are cut only at **camera shot boundaries** and never mid-line, so every clip can actually be cut.
- "Hook in the first 3s" is judged from dialogue, action and SFX. I can't see the frames, so visual-only hooks get less credit.
- Multi-segment promos stay within one episode: following episode N's cliffhanger with episode N+1 *resolves* it. Instead I built the **flash-forward** format common in microdrama ads: the climax beat first, then the setup, cutting just before the climax lands again.
- New viewers land on episode 1, so for posting order, early-episode promos beat late ones of the same quality (faster payoff, fewer spoilers).

2 How it works

Parse (50 scripts → 3,727 timed events) › **Cut** (14,291 clips, 20–60s, shot-aligned + flash-fwd) › **Score (rules)** (hook · cliffhanger · standalone · intensity) › **Select** (best 2 non-overlapping per episode) › **Judge (LLM)** (cold-viewer re-score of the top 40) › **Rank** (quality × funnel → report + editor sheet)

Kept as code / rules	Given to the AI	Human in the loop
Parsing, cut points, never splitting a line, 20–60s bounds, de-duplication, posting order. These must be exact. Scoring uses the script's own emotion tags, e.g. `(Cold, Commanding)`, plus small word lists for shocks, reveals and threats.	The judgement rules can't make: *"would a stranger understand this?"* The LLM sees **only the clip's transcript**, with no show context. It returns scores and quotes the hook and cliffhanger. A **guardrail rejects any quote not actually in the clip.**	The editor gets a keep/trim/reject sheet for 25 clips instead of watching about 75 minutes of episodes. Clips with a quality flag are marked for a human check.

Swappable model: one flag, `--llm gemini | openai | anthropic | none`. The `openai` mode takes any OpenAI-compatible URL (Groq, OpenRouter, Ollama, or an API Kuku provides). Plain HTTPS with no SDKs, a model fallback chain, and cached responses. With no key, rules-only still gives the full output.

3 Results: what to post first (rules + Gemini 3.5 Flash, all 50 episodes)

#	Clip	Hook (first 3 s)	Cliffhanger (cut point)	Why post it first
1	Ep18 0:56–1:29 · 33s	"The ledger!" as she runs back into a burning house	Beam crashes where she stood	Pure action, zero context needed, ends mid-disaster
2	Ep15 0:44–1:26 · 42s	"Get out of the way!" then she's shoved	"You want to fight us, veteran?"	Cuts right before the fight
3	Ep5 flash-fwd · 33s	Mother's ledger flung into the mud	"Then you should have guarded it better."	Shows the cruelty first, then makes you wait
4	Ep1 0:39–1:21 · 42s	"Because he is marrying me."	"Father..."	Jilted at the altar, and it's episode 1
5	Ep13 0:24–0:56 · 32s	"I would rather face the gallows than watch you hurt yourself"	"No..."	Self-sacrifice plus a clear romantic stake

What the LLM changed. The rules and Gemini agree on the top 3 by quality (quality ranks used below). Gemini pulled 5 clips into the top 25 that the rules had missed, moved Ep24's "Do you want the same punishment?" from #25 to #4, and dropped Ep7 from #5 to #19 because its "opening line is too weak to stop a fast scroll", a call the rules couldn't make. It also found a better cut point for Ep11. **40 calls, 0 failures, 0 invented quotes.** The full top 25, the best clips for all 50 episodes, and every transcript are in the raw outputs. **Two orderings:** the table above is post-first order (quality × funnel weight, as in report.md). promos.csv is by quality

alone, so Ep28 and Ep45 rank higher there; they drop to #6 and #20 here because of a spoiler-risk flag. The top 25 covers 21 episodes; best_per_episode.csv covers every episode.

4 How I know it's good, and where it fails

Automatic checks on every output clip, printed in report.md: length 20–60s, no cut inside a line, a real hook in the first 3s, minimum hook and cliffhanger scores, plus warnings for speaker count, scene changes and spoiler zone. All 25 pass the length, mid-line and first-3s checks; #23 (Ep46) fails the minimum cliffhanger score, and 16 of 25 carry at least one flag. I read every transcript myself. **Where it picks badly** (all from real runs):

- **Words, not meaning.** A cosy hearth scene ("dry wood catching fire") scored as danger. Ep7's second pick ends on a grateful smile, which resolves instead of leaving a cliffhanger.
- **Visual reveals are underweighted.** Ep50's real cliffhanger, Agnes silently witnessing the affair, is a camera move with no dialogue, so it ranked low.
- **Same character, different names.** Alexander is also "MYSTERIOUS BEARDED MAN" and Clarissa is also "LADY IN GOLD GOWN", which inflates the speaker count.
- **My spoiler rule may be wrong here.** The title already says "He was a Duke", so Ep42's reveal is probably the *best* ad. Spoiler weighting should be set per show.
- **The LLM rewards drama over clarity.** It promoted Ep45 and Ep46 despite rating them only 5–6/10 for making sense on their own.
- **No ground truth.** I don't have the promos Kuku actually posted, so the weights (rules 40% / LLM 60%) are hand-tuned, not learned.

5 Scale and cost

- **Rules only:** about 12 s of CPU per 50-episode show, $0, deterministic. It can run as a batch on every show at upload.
- **With an LLM:** 40 calls, roughly 50k tokens per show, which is cents on a flash-tier model. Re-runs are free (cached). The free tier took about 15 minutes because of rate limits.
- Only the shortlist goes to the LLM. Sending all 14k clips would cost about **350× more** for little gain.
- New genre: edit the word lists in `score.py`. New script format: write a new parser. Nothing else changes.

6 How I'd know it worked

- **Week 1:** editor keep rate on the review sheet (target ≥ 60%) and minutes saved per show.
- **After posting:** 3-second hold rate measures the hook; completion rate measures pacing; click/install rate measures the cliffhanger.
- Feed kept/rejected labels and ad results back in to **learn the weights** instead of hand-tuning them.

7 AI tools I used, and where they got it wrong

Tools: Claude Code (Anthropic's coding assistant) wrote most of the Python from my plain-English instructions. Gemini 3.5 Flash on the free tier is the judge that re-scores the shortlist. The useful part was how fast I could try an idea, read the clips it produced, and throw it away if it didn't hold up. **Where it got it wrong:** the first flash-forward version started the teaser on the gasp *after* the riding-crop strike (Ep24), so the strike also appeared in the setup and the "cliffhanger" repeated the payoff. I caught it by reading the transcripts and fixed it by starting the teaser at the first beat. Two more: an invented LLM quote still replaced the real one after being flagged, and the AI hard-coded a Gemini model Google had already retired. Both are fixed.